

ECE 3355: Electronics

Spring 2025

Exam 1

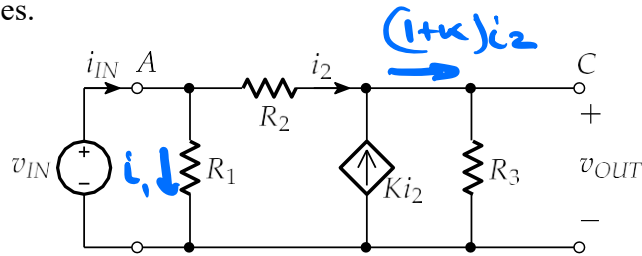
March 1, 2025

Do not open the exam until instructed to do so. Complete the exam on your own, without the help of your notes, prior examples or solutions, your book, or any communication/interaction with others. You must write a complete solution that shows the steps you took to solve the problems to receive full credit. You may use your calculator and a crib sheet provided as part of this exam. Answer the questions in the spaces provided on the question sheets. If you run out of room for an answer, continue on the back of the page. **You will have 1 hour and 15 minutes to finish the exam.**

Student's Name: _____

Question	Points	Score
1	20	
2	20	
3	20	
4	20	
5	20	
Total:	100	

1. (20 points) For the following circuit, find the voltage gain, $\frac{v_{OUT}}{v_{IN}}$, and the input and output resistances.

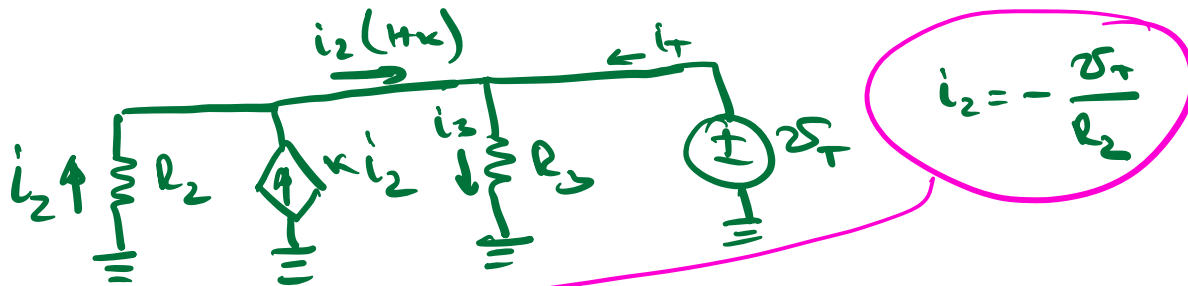


$$\left. \begin{aligned} v_{out} &= (1+K)i_2 R_3 \\ v_{in} &= i_2 R_2 + (1+K)i_2 R_3 \end{aligned} \right\} \frac{v_{out}}{v_{in}} = \frac{(1+K)R_3}{R_2 + (1+K)R_3}$$

$$R_{in} = \frac{v_{in}}{i_{in}} ; \quad i_{in} = i_1 + i_2 = \frac{v_{in}}{R_1} + i_2$$

$$R_{in} = \frac{\cancel{i_2 R_2} + (1+K)\cancel{i_2 R_3}}{\cancel{i_2 R_2} + (1+K)\cancel{i_2 R_3} + \cancel{i_2} R_1} = \frac{R_2 + (1+K)R_3}{R_2 + (1+K)R_3 + R_1} \cdot R_1$$

$$R_{in} = [R_2 + (1+K)R_3] \parallel R_1$$

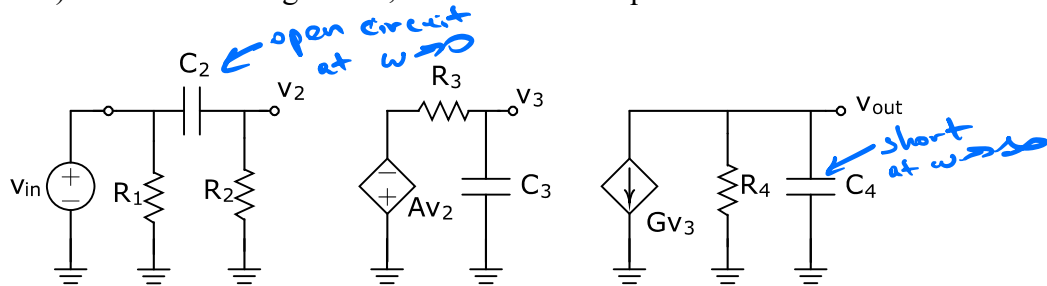


$$i_3 = \frac{v_T}{R_3} = i_2(1+K) + i_T$$

$$\frac{v_T}{R_3} = -\frac{v_T}{R_2}(1+K) + i_T \quad i_T = v_T \left[\frac{1}{R_3} + \frac{1+K}{R_2} \right]$$

$$R_{out} = \frac{v_T}{i_T} = \frac{1}{\frac{1}{R_3} + \frac{1+K}{R_2}} = R_3 \parallel \frac{R_2}{1+K}$$

2. (2 points) For the following circuit, answer the three questions below.



- (a) What is the gain at low frequencies?

(a) 0

- (b) What is the gain at high frequencies?

(b) 0

- (c) Derive the expression for the transfer function $\frac{\bar{V}_{out}}{\bar{V}_{in}}$ and present it in the form that allows easy identification of poles and zeros.

$$v_2 = \frac{R_2}{R_2 + \frac{1}{j\omega C_2}} v_{in} = \frac{j\omega C_2 R_2}{1 + j\omega C_2 R_2} v_{in}$$

$$v_3 = -\frac{\frac{1}{j\omega C_3}}{R_3 + \frac{1}{j\omega C_3}} A v_2 = -\frac{1}{1 + j\omega C_3 R_3} A v_2$$

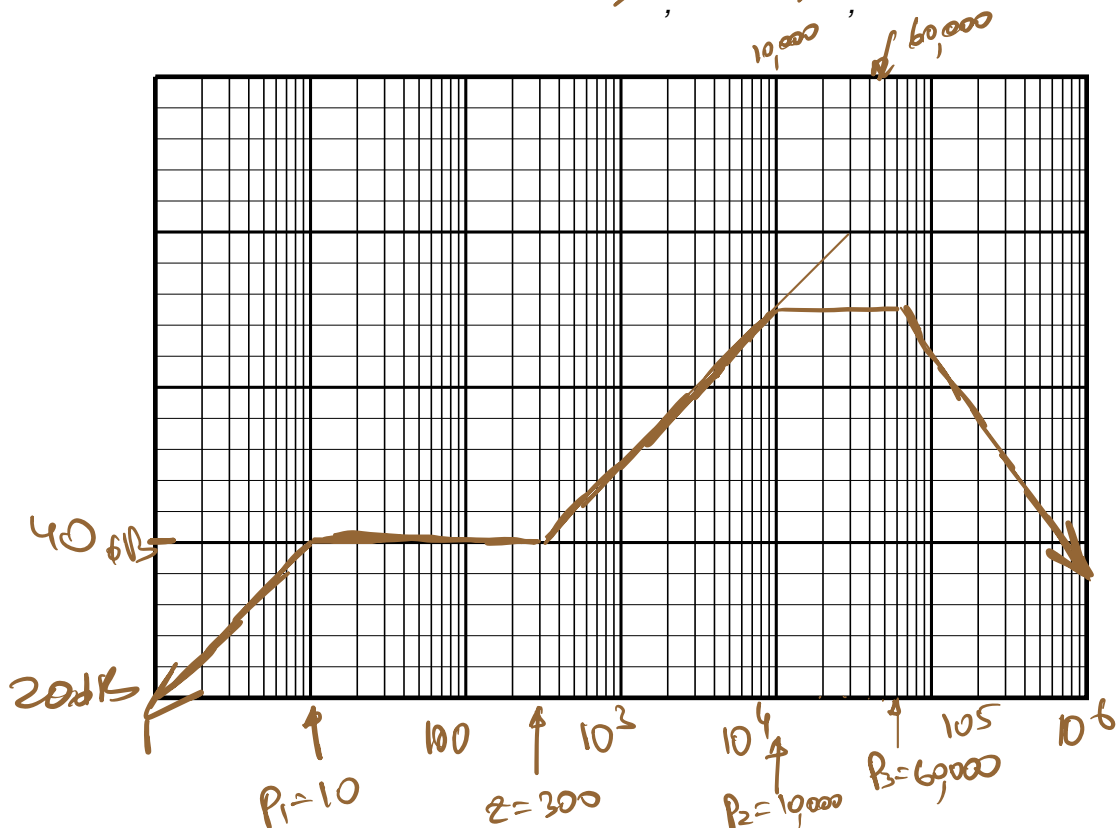
$$v_{out} = -G v_3 \frac{R_4 \cdot \frac{1}{j\omega C_4}}{R_4 + \frac{1}{j\omega C_4}} = -\frac{R_4 G}{1 + j\omega C_4 R_4} v_3$$

$$\frac{v_{out}}{v_{in}} = \frac{j C_2 R_2 A G R_4 \omega}{(1 + j\omega C_2 R_2)(1 + j\omega C_3 R_3)(1 + j\omega C_4 R_4)}$$

3. (20 points) Bode Plots

- (a) Plot a straight-line approximation of the Bode plot on the graph paper provided for the *magnitude only* for the following transfer function (the units for the values in the numerator and denominator of the imaginary terms is rad/sec).

$$H(j\omega) = \frac{10 (1 + j\omega/300)}{(1 + j\omega/10)(1 + j\omega/10000)(1 + j\omega/60000)}$$



- (b) What is the starting phase (ie., the phase at low frequencies)?

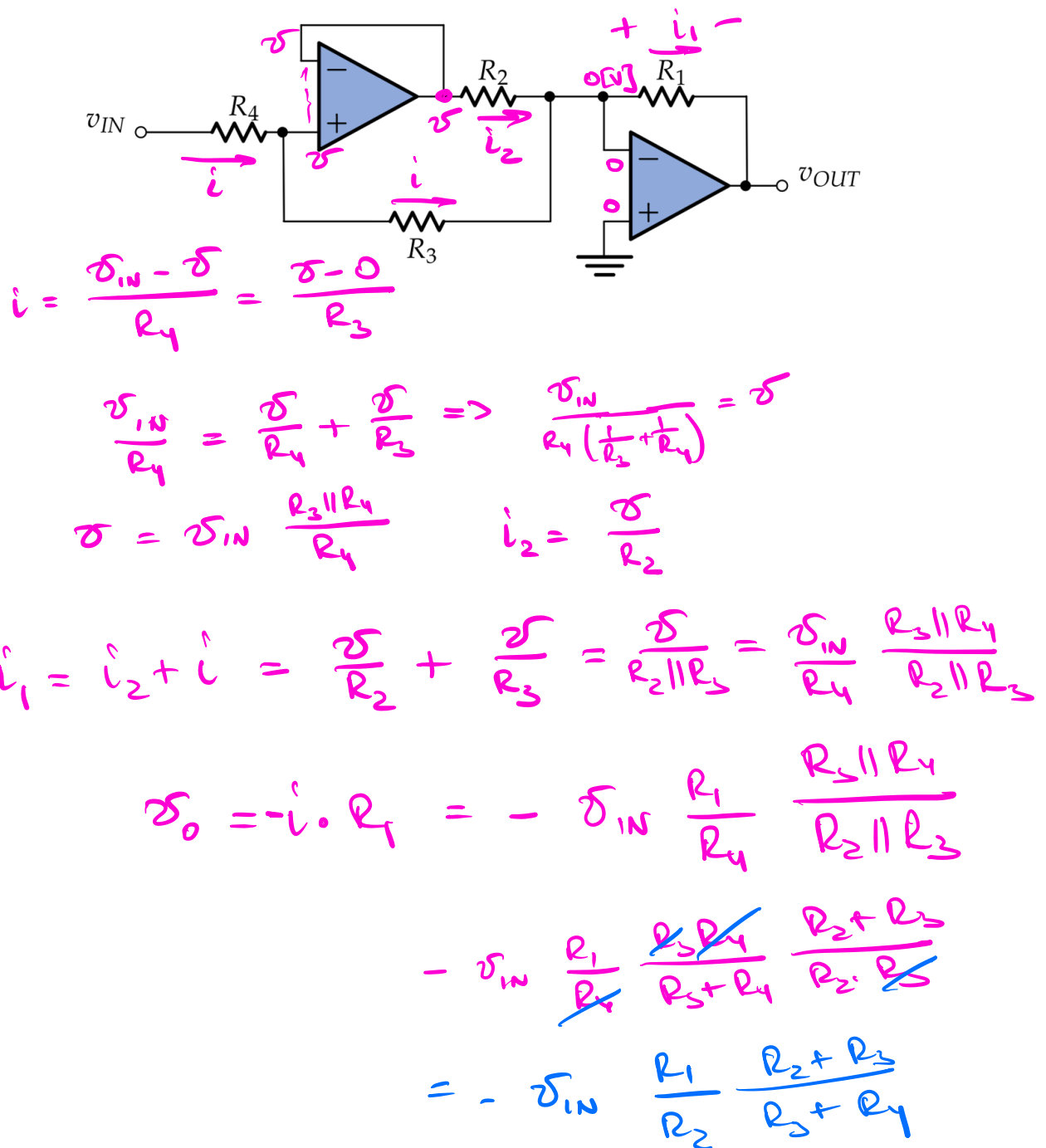
$$90^\circ - 90^\circ + 90^\circ - 90^\circ - 90^\circ$$

(b) 90

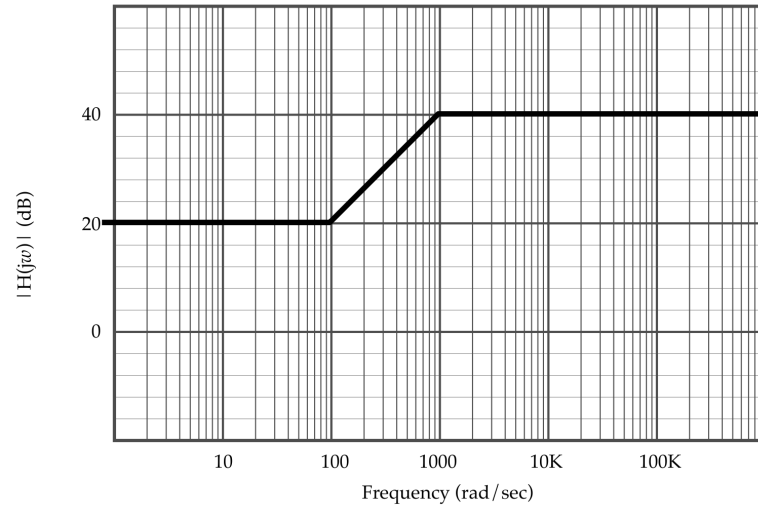
- (c) What is the ending phase (ie., the phase at high frequencies)?

(c) -90

4. (20 points) For the following circuit, find an expression for the voltage gain,



5. (20 points) A customer asks you to design a circuit that has the following frequency response (a straight-line approximation of its magnitude) using only ideal op-amps, capacitors (any value), and resistors (any value). The circuit should also not be affected by the source resistance.



$$\frac{V_{out}}{V_{in}} = \frac{1 + j\omega(R_1 + R_2)C_1}{1 + j\omega R_1 C_1}$$



$$1 + \frac{R_2}{R_1} = 10$$

$$R_2 = 9k$$

$$R_1 = 1k$$

$$\frac{1}{R_1 C_1} = 1000$$

$$C_1 = \frac{1}{1000 \cdot R_1} = \frac{1}{10^6} = 1\mu F$$

$$1 + \frac{R_4}{R_3} = 10$$

$$R_3 = 1k$$

$$R_4 = 9k$$